

2 Regulation with Noisy Signalling

We now turn to our main focus, the DMV's signalling problem. Can the DMV cheat? What if the DMV gives the seller a noisy signal? Realistically, the DMV's signal can have some error in it. Problems might arise in the inspection or they might do a poor job at inspecting the car. Suppose the DMV is profit-motivated and seeks to maximize its profit, with welfare being only a secondary concern. It may also be costly for the DMV to do a full inspection of the car to determine its true type θ . Instead, they might shirk and do a low level inspection, revealing a noisy signal, or maybe they might even try to give the seller a fake signal completely. This is likely not beneficial for social welfare. Can the DMV still prevent the market failure with a lower quality signal and get away with it, achieving a pareto efficient outcome? Can the DMV even improve on their total profit (more than their profit from sending the true θ signal)? These are some interesting questions that motivate our study of the DMV's signalling problem.

We begin by stating several possible information structures and discussing them briefly. First signal structure: The DMV gives a signal $\theta + \varepsilon$ where ε is drawn from a distribution with cdf F and density centered at 0. This model is realistic, for example, letting ε be standard normal noise. This signal can be interpreted as the DMV attempting to perform an accurate inspection of the car, but the result may have some error in it.

Second signal structure: DMV gives θ with probability p and gives an uninformative signal with probability $1 - p$, possibly just drawn randomly from $[0, 1]$, ie. $s \sim U(0, 1)$. This model is interesting as there is again noise in the signal, but unlike the first signal structure, the signal is not centered around θ . Here the DMV is choosing a lottery over two signal structures - giving a perfect signal ($s = \theta$) and an uninformative signal ($s \sim U(0, 1)$ independent of θ). It may not be clear why such a signal structure may be beneficial to the DMV compared to a perfect signal ($p = 1$), but it turns out that the randomized arbitrary signal in the $s \sim U(0, 1)$ is not as innocuous as upon first glance. Given the particular market structure and how the buyers view different car sellers under the information asymmetry, the inclusion of this noise can create an interesting effect on the market. This signal structure

leads to new considerations for sellers in their decision to disclose or not. In particular, lower seller types may have greater incentive to disclose. This signal structure will be the main topic of this paper.

Third signal structure: DMV gives a signal drawn from some distribution G with support containing $[0, 1]$ independent of θ . This is a totally uninformative signal. Such a signal can be interpreted as the DMV does not perform any inspection of the car at all and completely fabricates an arbitrary quality for the seller to display to the buyer. Notice that this is equivalent to the $p = 0$ signal of the second. As expected, this cannot be sustained. However, to gain intuition about the model and the signal structure, we will begin with this “completely fake” signal which provides no information about the true car quality and discuss how the market responds to it formally.

Let us consider how the market responds to the signal $s \sim U(0, 1)$ regardless of what θ is. Under such a completely uninformative signal, a buyer that sees a seller without displayed quality is willing to pay for $E(\theta \mid \text{undisclosed})$, whereas if he sees some seller with disclosed quality s he is willing to pay for

$$E(\theta \mid \text{disclosed}, s) = E(\theta \mid \text{disclosed})$$

as s provides no information about the car’s actual quality. Given this, a seller (of any type) will choose to disclose if

$$E(\theta \mid \text{disclosed}) - c \geq E(\theta \mid \text{undisclosed})$$

Since this decision criteria is the same for all seller types (given the decisions of the other sellers to disclose or not), it follows that the sellers must either all disclose or all undisclosed. The latter is not a reasonable equilibrium given that no information is conveyed to the buyer (as there is no nontrivial partition into disclose/undisclosed) and all sellers suffer a cost of c , so we conclude that no types will disclose, as expected. Thus, the market failure is not prevented.

Although this simple analysis yields the trivial result that we expected - no one will pay

for a completely uninformative signal - it does highlight the central incentive for a seller to disclose. There is a tradeoff between the cost incurred for disclosing by purchasing the DMV's certificate and the strategic "cost" of having the quality hidden from the buyer. Intuitively, if the signal is informative, then higher types will disclose while lower types will not disclose - even if the signal is noisy. Thus, if a seller decides to not disclose the car quality, her perceived quality to the buyer may be lower as she will be grouped with other, lower quality sellers that also do not disclose.

Now we turn to our central focus, the second signalling information structure, where the DMV charges c and gives θ with probability $p > 0$ and with probability $1 - p$ gives random $r \sim U(0, 1)$ drawn uniformly, independent of θ . We say that this is the DMV's signal s . Then the buyer is willing to pay for:

- $w = E(\theta \mid s = \emptyset) = E(\theta \mid \text{undisclosed}) = \hat{\theta}$ if θ is not disclosed.
- $w = p\theta + (1 - p)E(\theta \mid \text{disclosed})$ if $s = \theta$ is disclosed, because if the disclosed value is θ , with probability p it is the true θ and with probability $1 - p$ it is a fake signal and in that case the expected value of θ is $E(\theta \mid \text{disclosed})$. More formally, if the buyer observes signal s , he is willing to pay for

$$\begin{aligned} E(\theta \mid s) &= \Pr(\text{true signal})E(\theta \mid s, \text{true signal}) + \Pr(\text{fake signal})E(\theta \mid s, \text{fake signal}) \\ &= p \cdot s + (1 - p) \cdot E(\theta \mid \text{disclosed}) \end{aligned}$$

by total expectation.

We also have by total expectation

$$\begin{aligned} E(\theta) &= \frac{1}{2} = \Pr(\text{undisclosed})E(\theta \mid \text{undisclosed}) + \Pr(\text{disclosed})E(\theta \mid \text{disclosed}) \\ &= q\hat{\theta} + (1 - q)E(\theta \mid \text{disclosed}) \end{aligned}$$

And so $E(\theta \mid \text{disclosed}) = \frac{\frac{1}{2} - q\hat{\theta}}{1 - q}$ where q is the probability θ is not disclosed.

Let $w(\theta)$ be the buyer's willingness to pay if he sees a seller with disclosed quality θ . Then we have

$$w(\theta) = p\theta + (1-p)E(\theta \mid \text{disclosed}) = p\theta + (1-p) \cdot \frac{\frac{1}{2} - q\hat{\theta}}{1-q}$$

Let $W(\theta_0)$ then be the expected profit (buyer's expected willingness to pay) for seller of type θ_0 if she discloses. With probability p the θ_0 is accurately displayed to the buyer, in which case the buyer is willing to pay $w(\theta_0)$, and with probability $1-p$ some random $\theta \sim U(0,1)$ is drawn uniformly in which case the buyer is willing to pay $w(\theta)$.

Thus we have

$$\begin{aligned} W(\theta_0) &= pw(\theta_0) + (1-p) \int_0^1 w(\theta) d\theta \\ &= p \left(p\theta_0 + (1-p) \cdot \frac{\frac{1}{2} - q\hat{\theta}}{1-q} \right) + (1-p) \int_0^1 \left(p\theta + (1-p) \cdot \frac{\frac{1}{2} - q\hat{\theta}}{1-q} \right) d\theta \end{aligned}$$

So type θ_0 will

- disclose if

$$W(\theta_0) - c > \hat{\theta}$$

- not disclose if

$$W(\theta_0) - c \leq \hat{\theta}$$

Clearly $W(\cdot)$ is strictly increasing in θ_0 as $\frac{\partial W}{\partial \theta_0} = p^2 > 0$

It follows that the car dealers still employ a cutoff strategy with $\theta \in [0, q]$ not disclosing, and so $\hat{\theta} = \frac{q}{2}$, which means

$$E(\theta \mid \text{disclosed}) = \frac{\frac{1}{2} - q\hat{\theta}}{1-q} = \frac{1+q}{2} = \frac{1+2\hat{\theta}}{2}.$$

Thus

$$\begin{aligned}
W(\theta_0) &= p \left(p\theta_0 + (1-p)\frac{1+2\hat{\theta}}{2} \right) + (1-p) \int_0^1 \left(p\theta + (1-p)\frac{1+2\hat{\theta}}{2} \right) d\theta \\
&= (1-p)\frac{1+2\hat{\theta}}{2} + p \cdot p\theta_0 + (1-p) \int_0^1 p\theta d\theta \\
&= (1-p)\frac{1+2\hat{\theta}}{2} + p^2\theta_0 + (1-p)\frac{p}{2} \\
&= (1-p)\frac{1+p+2\hat{\theta}}{2} + p^2\theta_0
\end{aligned}$$

Now solving:

$$(1-p)\frac{1+p+2\hat{\theta}}{2} + p^2\theta_0 - c > \hat{\theta}$$

ie.

$$(1-p)(1+p+2\hat{\theta}) + 2p^2\theta_0 - 2c > 2\hat{\theta}$$

$$1 - p^2 + 2p^2\theta_0 - 2c > p \cdot 2\hat{\theta}$$

$$2p^2\theta_0 > p \cdot 2\hat{\theta} + p^2 - 1 + 2c$$

$$\theta_0 > \frac{p \cdot 2\hat{\theta} + p^2 - 1 + 2c}{2p^2}$$

So the types that don't disclose are:

$$\theta \leq \frac{p \cdot 2\hat{\theta} + p^2 - 1 + 2c}{2p^2}$$

And so we have

$$\hat{\theta} = E \left(\theta \mid \theta \leq \frac{p \cdot 2\hat{\theta} + p^2 - 1 + 2c}{2p^2} \right) = \min \left(\frac{p \cdot 2\hat{\theta} + p^2 - 1 + 2c}{4p^2}, \frac{1}{2} \right)$$

Solving:

$$4p^2\hat{\theta} = p \cdot 2\hat{\theta} + p^2 - 1 + 2c$$

$$2p(1-2p)\hat{\theta} = 1 - p^2 - 2c$$